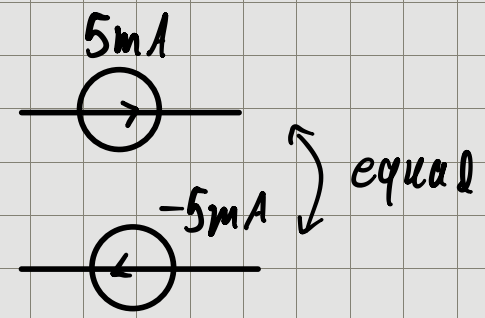
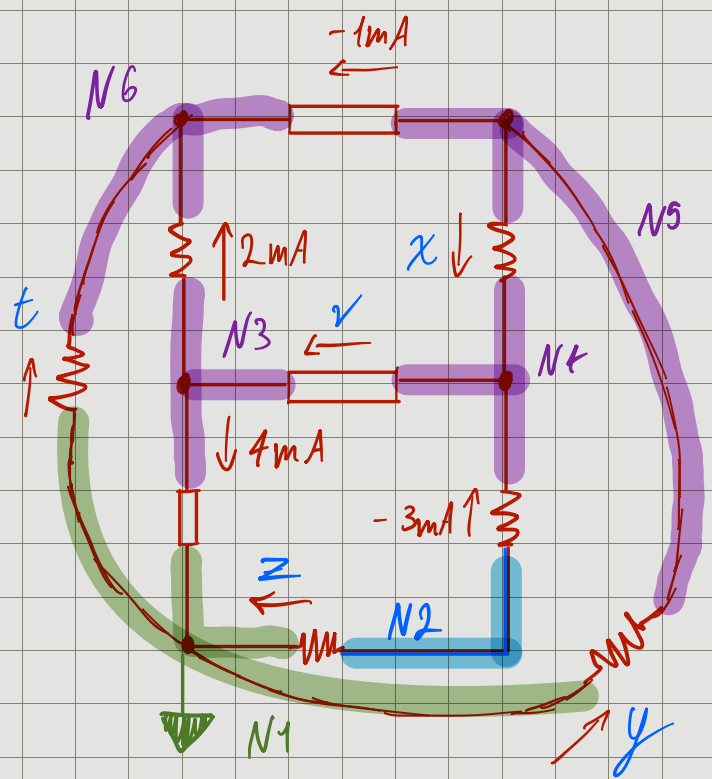
6. Kirchoff's Current Law

The sum of all currents flowing into a node equals the sum of all currents flowing out of the same node

$$\sum I_{in} = \sum I_{out}$$





Find t, v, x, y, z

Nodes = 6

KCL on N1:

$z + 4m = t + y$

$7m = t + y$

KCL on N2:

$z - 3m = 0$

$z = 3m$

z

KCL on N3:

$v = 2m + 4m$

$v = 6m$

v

KCL on N4:

$x - 3 = v$

$x - 3m = 6m$

$x = 9m$

x

KCL on N5:

$y = x - 1m$

$y = 8m$

Agrees y

KCL on N6:

$t + 2m - 1m = 0$

$t = -1m$

t

7 KVL: Kirchoff's Voltage Law

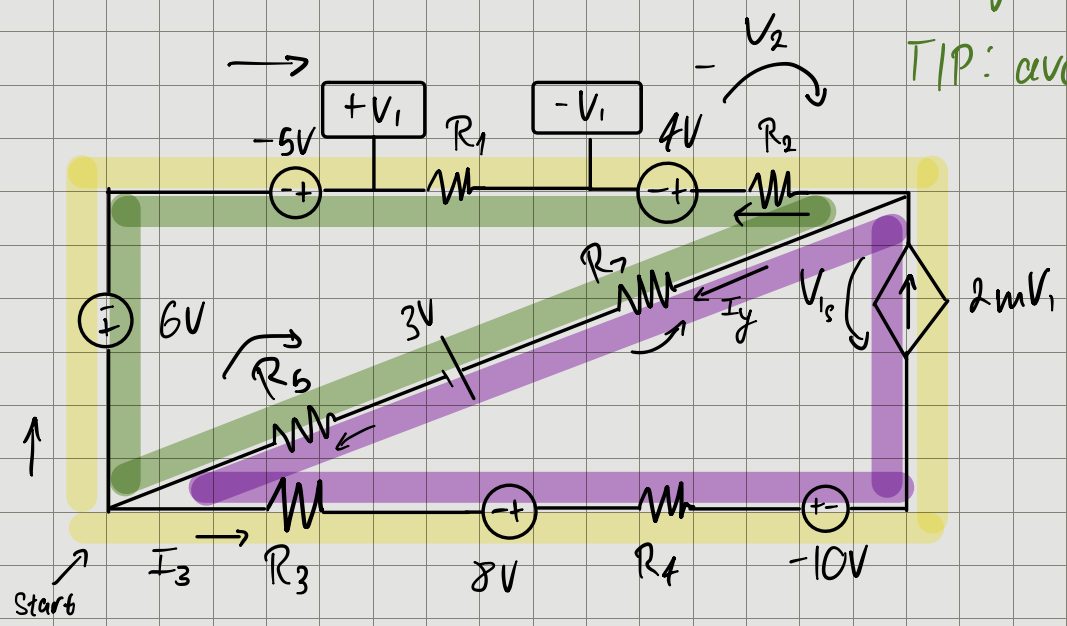
The sum of all voltage drops around a loop is zero.
A loop is any closed sequence of branches and nodes.

$$\Delta V = 0$$

How to do KVL for first midterm exam

Start on bottom left and go around the loop clockwise, using the same rules as for voltage drops.

TIP: avoid current supplies. They also have voltage drops!



Question: Write all KVLs for the circuit:

KVL on Green loop:

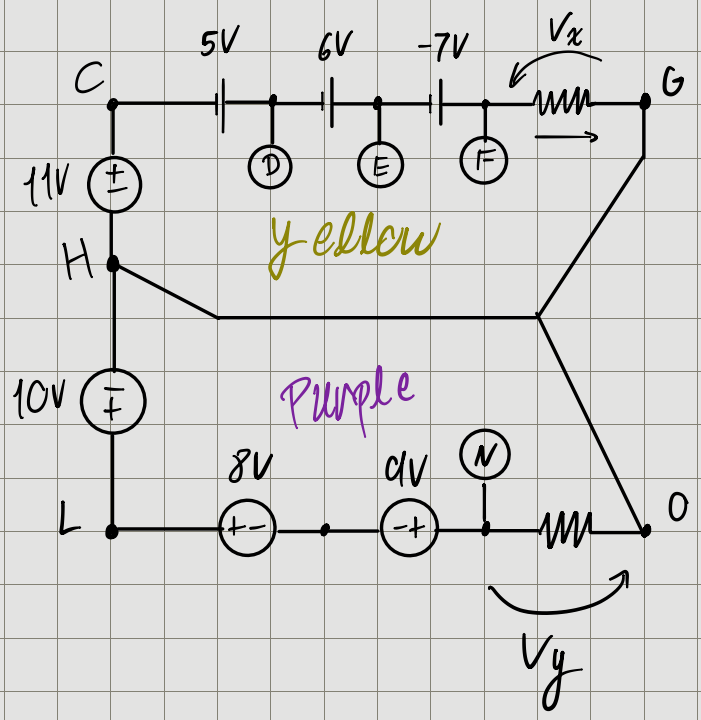
$$6-(-5)+V_1-4-V_2+V_7+3+V_5$$
$$10+V_1-V_2+V_7+V_5=0$$

KVL on Purple loop:

$$-V_5-3-V_7-V_{I_s}+10+V_4+8-V_3$$
$$-V_5-V_7-V_{I_s}+V_4-V_3+15=0$$

KVL on yellow loop:

$$11+V_1-4-V_2-V_{I_s}+10+V_4+8-V_3$$
$$25+V_1-V_2-V_{I_s}+V_4-V_3=0$$



KVL yellow:

$$-11-5-6+7+V_x=0$$

$$V_x=15V$$

KVL purple:

$$10+V_y+9-8=0$$

$$V_y=-11V$$